

Causal Inference Assignment

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1 Introduction

Experiments have long been the exclusive home of causal inferences due to their randomized treatment assignment. However, in the world of social sciences, and many other fields, experiments aren't always possible to run. The science of making causal inferences from observational studies has developed tremendously over the past 50 years (see chapter 2 of Imbens & Rubin (2015)). We'd summarize the biggest finding as this; for observational study participants with similar covariates, given those covariates are sufficient to remove confounding, causal effects can be inferred (see, for example, chapter 4 of Huber (2023), or chapter 3 in Pearl et al. (2016)).

The aim of this report is to do just that, to infer a causal effect from (a simulation of) an observational study. Specifically, we want to compare the causal risk difference (defined as $E[Y(1) - Y(0)]$) of smoking on severe COVID-19. The data we will use to do so are from a simulation study by Reinhammar & Waernbaum (2025), where the "true" risk difference is defined as 0.035 [6.1]. We will estimate this risk difference with regression imputation using three different methods; linear-, logistic- and kernel regression.

The analysis is conducted on three different data sets (A, B, and C), with each representing a different underlying model for generating the potential outcomes. The report will begin by providing some more detail on these and the rest of the data. After that, to address potential confounding, we identify a sufficient adjustment set by applying d-separation and the backdoor criterion to the study's Directed Acyclic Graph (DAG) [6.2]. The third section covers the three regression imputation techniques we will be using, the fourth presents the results of each estimator in each scenario, and in the fifth, we conclude.

2 Data

The data we will be analysing are taken from a simulation study conducted by Reinhammar & Waernbaum (2025). In this study, the authors use a data learner (called `mscovid_sim`) to imitate the data set from Louapre et al. (2020) - an observational study investigating risk factors for COVID-19 severity for patients with multiple sclerosis. Reinhammar & Waernbaum (2025) created a DAG of the hypothesized underlying causal structure, which we have attached in the appendix [6.2].

There are three data sets, each with 3,470 units, and 15 variables. The only difference between them is how the outcome variable is generated. The primary variables of interest and the outcome models are:

- **Outcome (Y):** Severe COVID-19, a binary variable which equals one if the unit has a severity score of 3 or more on a 7-point scale.
- **Treatment (D):** Smoking status, modeled as a binary variable (one when the unit is a smoker).
- **Covariates (X):** The data set includes clinical and demographic characteristics such as Age, Sex, BMI, diabetes status, etc... All covariates can be found in the DAG [6.2]
- **Outcome model A:** Generated using a logistic model with only linear terms.
- **Outcome model B:** Generated using a logistic model that includes interactions between the treatment and the covariates.
- **Outcome model C:** Generated using a probit model featuring non-linear terms and interactions both between covariates and between the treatment and covariates.

In terms of assumptions; Reinhammar & Waernbaum (2025) state that data was generated so that the SUTVA is met. Given the authors also defined a DAG, we can use d-separation and the backdoor criterion to form a sufficient adjustment set (Reinhammar and Waernbaum, 2025, p.6).

2.1 The sufficient adjustment set

Having presented our data, we shift our attention to reducing our set of covariates to a sufficient adjustment set. The idea of this set is directly tied to the assumption of unconfoundedness. In observational studies, this set must be conditioned on in order to make unbiased causal effect estimates - an idea covered in chapter 4 of Huber (2023). As mentioned above, a sufficient adjustment set in the presence of a fully defined DAG, can be found by a set of confounder variables that satisfies the backdoor criterion.

There are several interesting deductions about (in)dependence and conditional (in)dependence that can be found by analysing a DAG. Pearl et al. (2016) cover this in chapters 2 and 3 and we will deploy the d-separation technique (section 2.4) when checking the backdoor criterion (section 3.3) for "Smoking" and "COVID-19 Severity" on our relevant DAG [6.2]. The d-separation process involves analysing paths between nodes and finding which ones "block" paths between other nodes, rendering them independent.

With reference to the DAG mentioned, we see that Smoking, our treatment variable, has two arrows pointing towards it. These "parent" variables (Pearl et al., 2016) of "Sex" and "Age" are of special interest to us because (1) smoking is a descendant of both of them, and (2) there is an open path directly from Smoking, to them, straight to COVID-19 Severity. Without adjusting for Age and Sex, with smoking as the treatment, we will muddy our estimate of the ATE of Smoking on COVID-19 Severity as we have two confounding variables stopping us from isolating the effect/dependent relationship of interest.

By conditioning on both of these variables, Age and Sex, we account for this confoundedness and have thus formed our sufficient adjustment set. We check our set against Pearl et al.'s (2016) definition (3.3.1) of the backdoor criterion: '*A set of variables Z satisfies the backdoor criterion relative to (X, Y) if no node in Z is a descendant of X and Z blocks every path between X and Y that contains an arrow into X* '. It is clear from the DAG that Age and Sex are not descendants of Smoking. And, by blocking Age, and Sex, there are no open paths between Smoking and COVID-19 Severity that contain an arrow into Smoking, there are now only paths that are descendants of Smoking. Hence, **{Age, Sex}** forms the sufficient adjustment set, and the two covariates will be included in the proceeding analysis.

3 Regressions imputation

Given our sufficient adjustment set, we can feel more confident about being able to infer causal effects between units with similar values for the covariates. It would be perfect if we could compare participants with themselves - no other participant can possibly have closer covariates than that. As this is impossible (Holland (1986) calls it the "fundamental problem of causal analysis"), the motivation for imputation arises. We will predict the missing potential outcome using linear-, logistic- and kernel regression.

To estimate the risk differences $E[Y(1) - Y(0)]$ (denoted Δ) for all three regression imputations, we use the following estimator, as seen in the discussion of section 4.2 of Huber (2023):

$$\hat{\Delta}_{reg} = \frac{1}{n} \sum_{i=1}^n [\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)] \quad (1)$$

Where $\hat{\mu}_1(X_i)$ is the estimator of the outcome variable obtained from the model "trained" on the smokers, and estimated using the full data set assuming every observation is a smoker (thus imputing the smoker values for the non-smokers). $\hat{\mu}_0(X_i)$ is the same but in the other direction. The difference between the observed and imputed value for a particular set of covariates ($\hat{\mu}_1(x_i) - \hat{\mu}_0(x_i)$) gives the conditional average treatment effect (the CATE). The average of the CATEs gives our risk difference estimate ($\hat{\Delta}_{reg}$).

For the linear regression model, to obtain $\hat{\mu}_1(X_i)$ and $\hat{\mu}_0(X_i)$, we regress the outcome Y (severity) on the two covariates (age and sex), where we estimate coefficients $(\hat{\alpha}_1, \hat{\beta}_{sex,1}, \hat{\beta}_{age,1})$ and $(\hat{\alpha}_0, \hat{\beta}_{sex,0}, \hat{\beta}_{age,0})$

for the treated (smokers) and non treated (non-smokers) respectively. The two model specifications are:

$$\hat{\mu}_1(X_i) = \hat{\alpha}_1 + \hat{\beta}_{age,1}X_{age,i} + \hat{\beta}_{sex,1}X_{sex,i} \quad (2)$$

$$\hat{\mu}_0(X_i) = \hat{\alpha}_0 + \hat{\beta}_{age,0}X_{age,i} + \hat{\beta}_{sex,0}X_{sex,i} \quad (3)$$

Huber (2023) argues that the number of higher order terms and interaction terms is a practical issue in regression specifications. A more complex model may increase high variance due to overfitting, and a too simple model may induce a bias due to a bad approximation of the true model. In our case, by splitting the data set into smokers and non-smokers, we automatically include the interactions between age and smoking, and sex and smoking. We believe these are important interactions to include as without them, we assume a constant risk difference across all ages and sex, which we consider unrealistic.

Plugging values for $X_{age,i}$ and $X_{sex,i}$ of one of our observations into both Equations 2 and 3, we can obtain a CATE ($\hat{\mu}_1(x_i) - \hat{\mu}_0(x_i)$). Averaging all the CATEs (Equation 1) estimates the ATE, $\hat{\Delta}_{ols}$.

Given the outcome is binary, the probability of severe COVID-19 may fall outside of the boundaries of 0 and 1 when fitting a linear model. Therefore, we also fit a logistic regression using the same model specifications as in Equations 2 and 3, to see how it performs. This model is implemented in a similar way to the linear one, fitting the regression where treated and non-treated are separated to obtain different coefficients for each group. To estimate the risk difference, $\hat{\mu}_1(X_i)$ is obtained from the logit model:

$$\hat{\mu}_1(X_i) = \frac{\exp(\hat{\alpha}_1 + \hat{\beta}_{age,1}X_{age,i} + \hat{\beta}_{sex,1}X_{sex,i})}{1 + \exp(\hat{\alpha}_1 + \hat{\beta}_{age,1}X_{age,i} + \hat{\beta}_{sex,1}X_{sex,i})} \quad (4)$$

Both $\hat{\mu}_1(X_i)$ and $\hat{\mu}_0(X_i)$ from the logistic model are later inserted in Equation 1 to obtain the estimated risk difference, $\hat{\Delta}_{log}$.

One limitation when predicting values based on coefficients is that they can be far off from values in our data set when using e.g. linear regression. Predictions based on such models may be quite poor when we want to estimate $\mu_1(X)$ and $\mu_0(X)$ in situations where the relationship between variables is more complex. To counter this, one can use local approaches which do not permit predicted values that are far away from our observed data. Instead we can estimate $\mu_1(X)$ using the kernel function ($\mathcal{K}$) which gives observations with covariate values closer to a given vector (or scalar) x larger weights. This is known as the local constant kernel regression. As in Huber (2023), the kernel estimator $\hat{\mu}_{1,h}(X_i)$ for the treated is:

$$\hat{\mu}_{1,h}(x) = \frac{\sum_{i:D_i=1}^n \mathcal{K}\left(\frac{X_i - x}{h}\right) \cdot Y_i}{\sum_{i:D_i=1}^n \mathcal{K}\left(\frac{X_i - x}{h}\right)} \quad (5)$$

Where $D_i = 1$ indicates treated units, $\mathcal{K}$ represents the kernel function, and h is the bandwidth. The kernel function determines how weightings within a local neighborhood of x are distributed (we will expand briefly on which we use at the end of this section). The bandwidth is a crucial parameter that determines the "width" of this local neighborhood around x . A small bandwidth may introduce high variance due to weighting over few observations (overfitting), and a large bandwidth may lead to more uniform weights across a wider range of observations, which can lead to high bias. To choose the optimal bandwidth (h_{opt}) for the data, leave-one-out-cross-validation (LOOCV) is used:

$$h_{opt} = \arg \min_{h \in \mathcal{H}} \sum_{i:D_i=1} [Y_i - \hat{\mu}_{1,-i,h}(X_i)]^2 \quad (6)$$

h_{opt} is selected as the h with the smallest MSE from the LOOCV for each covariate included in the model. $\hat{\mu}_1(X_i)$ is calculated as $\hat{\mu}_{1,h_{opt}}(X_i)$ including all observations. The same procedure is used to

obtain $\hat{\mu}_{0,h_{opt}}(X_i)$ for the non-treated. The risk difference is then calculated similar to Equation 1. To run the kernel regression the package `np` is utilized. To find the optimal bandwidth we use the function `npregbw`, where the optimisation method is based on the MSE and LOOCV (as seen in Equation 6), and the `regtype` is set to `lc` indicating the local constant (we find a constant estimate at points rather than a sloped line, which is called local regression). The specific kernel functions used are the defaults under the `npregbw` function; the Aitchison and Aitken function for "Sex" and a second-order gaussian function for "Age" (see the `ckertype`, `ckerorder` and `ukertype` arguments in the R-documentation by Hayfield & Racine (2020)). After the optimal bandwidth is found, the kernel model is fit using `npreg`.

4 Results

In this section, the estimated risk differences for data sets A, B, and C are presented. The risk difference $E[Y(1) - Y(0)]$ has been calculated using the three different estimation methods: linear regression, logistic regression, and the non-parametric kernel estimator. We can evaluate whether the estimated impact of smoking on COVID-19 severity remains consistent across different functional forms. The results for all three data sets and estimators are summarized in Table 1 below.

Estimator	data set A	data set B	data set C
OLS	0.0481	0.0493	-0.0104
Logistic	0.0484	0.0479	-0.0105
Kernel	0.0538	0.0485	-0.0096

Table 1: Estimated Risk Differences across data sets and Estimators

The risk difference estimates are relatively equal across the estimator methods. In data set A the kernel estimator predicts a slightly higher risk difference of 0.0538 compared to the OLS and the logistic. In data set B the risk differences are very close to each other, all estimating an increase in the risk of severe COVID-19 for smokers compared to non smokers. In data set C all models predict a decrease in the risk of severe COVID-19 for smokers compared to non smokers.

5 Conclusion

After choosing the sufficient set of variables using the d-separation criterion, fitting the different models, and obtaining estimates of the risk differences, the models all give similar results across each data set. The linear regression model manages to get the closest estimate of the true effect for data set A, the logistic model manages to get the closest estimate for data set B, and for data set C the kernel estimator obtained the closest match. This could be because data set A is an outcome model with only linear terms which a simpler model like linear regression manages to predict the outcome of better. Outcome model C is a model with non-linear terms and interactions between covariates as well as between the treatment and covariates, a localized method like the kernel estimator manages to capture more complex relationships. Although there are some variations between the three models, it is hard to say that any one performed better than the other, especially in regards to data set B and C.

None of the models successfully manage to estimate the true effect of 0.035 and all models predict a decreasing risk of severe COVID-19 for data set C. Theoretically the kernel estimator might be preferred due to the complex relationships in real life scenarios, however, since the convergence rate is slower than $\frac{1}{\sqrt{n}}$ discussed in Huber (2023) this might be a poor choice when the sample size is small. As mentioned earlier in this case the models all leave a lot to be desired, however, the linear regression was quite a lot

closer to the true risk difference than the kernel estimator in data set A, therefore, it should be preferred in that scenario. Ultimately, it is hard to deduce exactly how well the estimators performed without also calculating confidence intervals.

Other estimators could be considered for the estimation of treatment effects. One estimator that would be an attractive choice is a doubly robust estimator like the Augmented Inverse Probability Weighting (AIPW). It combines conditional mean outcomes and weighting using the propensity score. When correctly specified, it can reduce bias, increase efficiency, and remains consistent if either the propensity score or the conditional mean outcome models are misspecified, in contrast to regression estimators which rely on correct specifications of $\mu_1(X_i)$ and $\mu_0(X_i)$.

6 Appendix

6.1 Derivation of the “true” ATE

$$\begin{aligned} E[Y(1) - Y(0)] &= E[Y(1)] - E[Y(0)] \\ &= (P[Y(1) = 1] \times 1 + P[Y(1) = 0] \times 0) \\ &\quad - (P[Y(0) = 1] \times 1 + P[Y(0) = 0] \times 0) \\ &= P[Y(1) = 1] - P[Y(0) = 1] \\ &= 0.24 - 0.205 \\ &= 0.035 \end{aligned} \quad (\text{Reinhammar and Waernbaum, 2025, p.8})$$

6.2 The DAG

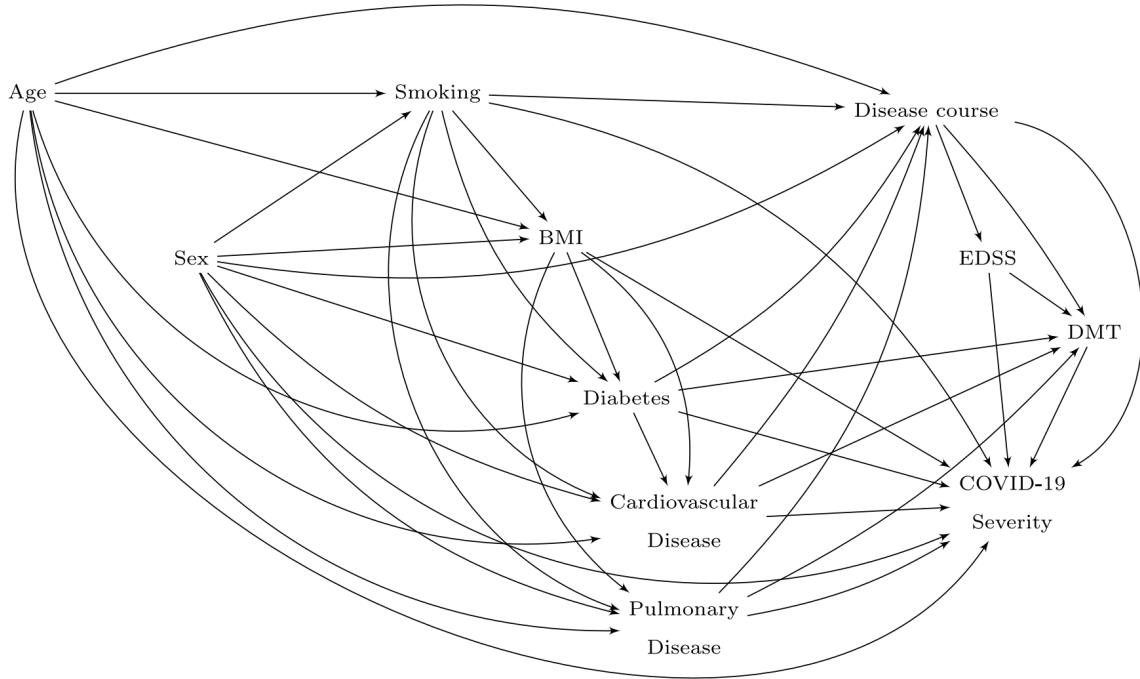


Figure 1: DAG displaying the data-generating model in the data learner

References

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